Exhausting the cubic matrix equations

Human Compression project: L023 working proof

This unreviewed component recovers, with explicit corrections, the polynomial construction and finite table in Leemans–Stokes, arXiv:2208.08215v2, Section 4. It does not settle the separate rank-four incidence assertions. The equations and existence construction belong to the source authors. The orbit exhaustion and the characteristic-two correction below are the present argument. No conjecture, whole-original, formalization or compression factor is claimed.

Put $K = \mathbb{F}_{q^3}$, $k = \mathbb{F}_q$, and let τ raise entries to their qth powers. Let $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The first three source equations are equivalent to

$$R = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \quad \det R = 1, \quad R^{\tau^2} = RR^{\tau}. \tag{1}$$

Here superscripts τ denote the field operation, not matrix powers. The source's trace equation is redundant in (1).

Lemma 1 (Odd characteristic)**.** *If q is odd and (1) has a solution, all its solutions form one free orbit under conjugation by* $\mathrm{PGL}_2(k)$*. There are exactly $q(q^2-1)$ solutions.*

Proof. Write $S = R^{\tau}$ and $T = R^{\tau^2} = RS$. Cayley–Hamilton gives $R^2 = S^2 = T^2 = -I$. Consequently $SR = -RS$. The matrices I, R, S, RS form a basis of $M_2(K)$: under the trace pairing their Gram matrix is diagonal with entries $2, -2, -2, -2$, all nonzero. Their multiplication is determined by $R^2 = S^2 = -I$ and $SR = -RS$. For any second solution R', the correspondence taking this basis to $I, R', S', R'S'$ is therefore a K-algebra automorphism of $M_2(K)$.

Every such algebra automorphism is conjugation by a matrix $C \in \mathrm{GL}_2(K)$. For completeness, choose a basis adapted to the images of the two orthogonal rank-one idempotents E_{11}, E_{22}. In that basis the images of E_{12}, E_{21} are respectively $uE_{12}, u^{-1}E_{21}$, by their products with the idempotents and each other. Rescaling one basis vector gives the claimed conjugation. Thus $R' = CRC^{-1}$ and $S' = CSC^{-1}$. Apply τ to these identities and compare with the identities for S', T'. The matrix $C^{-1}C^{\tau}$ commutes with S, T, hence with all of $M_2(K)$, and is scalar. Divide C by any one nonzero entry. All its entry ratios are fixed by τ, so this normalized matrix lies in $\mathrm{GL}_2(k)$. This proves transitivity under $\mathrm{PGL}_2(k)$. A projectivity stabilizing R also stabilizes S, hence is scalar by the same basis calculation. The action is free, giving the count. □

Lemma 2 (Characteristic two)**.** *If q is even, the nonscalar solutions of (1) are parametrized without repetition by a point of $\mathbb{P}^1(k)$ and a nonzero element of* $\ker \mathrm{Tr}_{K/k}$*. Their number is $(q+1)(q^2-1)$. The only additional solution is I.*

Proof. A nonscalar trace-zero determinant-one matrix is a nonidentity unipotent involution, with a unique fixed point on $\mathbb{P}^1(K)$. With $S = R^{\tau}$ the identity $(RS)^2 = I$ says that R, S commute. They are distinct, since $R = S$ would give $R^{\tau^2} = I$. Two commuting nonidentity unipotents in $\mathrm{SL}_2(K)$ have the same fixed point: after conjugating one to upper triangular form, its centralizer is the group of upper unitriangular matrices. Their shared fixed point is preserved by τ, hence belongs to $\mathbb{P}^1(k)$.

Choose, once for each such point P, a matrix $C_P \in \mathrm{GL}_2(k)$ carrying infinity to P. Every solution with this point has the unique form

$$R = C_P \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} C_P^{-1}, \qquad t \neq 0.$$

Equation (1) is now exactly $t^{q^2} = t + t^q$, the trace-zero condition. The trace map is a nonzero k-linear map from a three-dimensional space to k (on k it is the identity), so its kernel has q^2 elements. This gives all the stated parameters without repetition. The scalar case forces $R = I$, which does satisfy (1). □

Proposition 3 (Primitive eigenvalues)**.** *Let $Q = q^3 \geq 8$. Suppose R satisfies (1) and RJ has order $Q-1$ or $Q+1$ in $\mathrm{SL}_2(K)$. Its projective images, together with R^τ, generate $\mathrm{PSL}_2(K)$ and give a Wilson class III map. The three periods are*

$$\frac{Q-1}{\gcd(2, Q-1)} \quad \text{or} \quad \frac{Q+1}{\gcd(2, Q-1)}, \tag{2}$$

respectively.

Proof. The projective images r, s of R, R^τ are distinct commuting involutions, and τ cycles r, s, rs while fixing the image j of J. Let $d = \gcd(2, Q-1)$ and $m = (Q \pm 1)/d$. The order assumption makes $\langle r, j\rangle$ the full dihedral torus normalizer of order $2m$. By the retained Dickson classification, this is maximal for the cubic field sizes $Q \geq 8$ under consideration. Thus $L = \langle r, j, s\rangle$ is either that dihedral group or the whole projective special linear group.

The former alternative is impossible. If m is odd the dihedral group has no Klein four subgroup. If m is even, each Klein four subgroup contains its unique central involution. But L is τ-stable, and τ cycles the three involutions of $\langle r, s\rangle$, so cannot fix that characteristic central involution. This proves generation. In odd characteristic the projective order is half the indicated matrix order; in even characteristic it is unchanged, proving (2). Finally τ is a noninner field automorphism. A Wilson duality would combine with it to give a faithful S_3 of marked automorphisms; its commutator would die in the abelian outer automorphism group, contradicting the nontrivial field image of τ. Thus the map has class III. □

The statement uses a *primitive* $(Q \pm 1)$st root of unity in the source's fourth equation. Allowing roots of smaller order would not give the required matrix order. Indeed a determinant-one matrix has characteristic polynomial $x^2 - \mathrm{tr}(RJ)x + 1$; the trace $\xi + \xi^{-1}$ for a primitive root gives the two distinct eigenvalues ξ, ξ^{-1}. The printed ambient $\mathrm{PSL}_2(q)$ in the proof of source Lemma 5 must be $\mathrm{PSL}_2(q^3)$. The half-orders printed after its table apply only in odd characteristic.

The complete finite comparison

The whole required program `polynomial_table.g` works in GAP's exact finite fields. For odd q it finds one literal solution by scanning a, b and setting $c = (-1 - a^2)/b$, with $b \neq 0$. That scan is used only to find a seed; it need not exhaust all possible seeds. The program checks that a seed is found for every odd q in the table. It then conjugates this seed by the following complete list of projectivities over k:

$$\begin{pmatrix} x & y \\ 0 & 1 \end{pmatrix} \ (x \neq 0), \qquad \begin{pmatrix} x & y \\ 1 & z \end{pmatrix} \ (xz \neq y).$$

Their respective sizes are $q(q-1)$ and $q^2(q-1)$. Normalizing the first nonzero entry of the bottom row proves both completeness and uniqueness. The odd-characteristic lemma therefore proves exhaustion. For even q the program uses $C_\infty = I$ and $C_x = \left(\begin{smallmatrix} x & 1 \\ 1 & 0 \end{smallmatrix}\right)$ for $x \in k$, and every nonzero trace-zero parameter. The omitted scalar solution I has $IJ = J$ of order two, and contributes to neither column.

For each resulting matrix the program checks determinant one, trace zero, and the full matrix equation (1), rejects duplicate matrices, checks the proved total, and computes the exact order of RJ. The output records the seed by exponents of GAP's $Z(q^3)$, with -1 for zero. Thus the finite

counts use the complete program and finite-field arithmetic as a computational premise; they are not inferred from a sample or a rendered table. No optional geometry probe is a premise.

q	nonscalar solutions	$Q-1$	$Q+1$	printed $Q-1$	printed $Q+1$
3	24	0	24	0	24
4	75	24	24	12	12
5	120	48	36	48	36
7	336	96	96	96	96
8	567	216	144	162	108
9	720	192	192	192	192
11	1320	324	216	324	216
13	2184	432	576	432	576
16	4335	960	1920	840	1680
17	4896	1440	624	1440	624
19	6840	1020	1440	1020	1440
23	12144	2952	2160	2952	2160

In characteristic two, the excess counts are respectively $(12, 12)$, $(54, 36)$ and $(120, 240)$. In the complete census these are exactly the qualifying solutions with $bc = 0$. Subtracting those solutions reproduces the three printed rows. The source does not impose $bc \neq 0$; these are therefore additional solutions to its printed equations under the primitive-root interpretation. This observation identifies which solutions account for the numerical discrepancy, without claiming to know how the source authors' computation produced it.

Scope. This is a component of a prospective corrected recovery. The retained inputs are the stated torus-normalizer part of Dickson's classification, the projective automorphism description and Wilson's marked-map correspondence. The two exhaustion lemmas are proved here. The displayed finite values require the whole GAP program. No claim about full geometric correlations of the rank-four construction follows from these calculations.